ANSWER ALL QUESTIONS

1. A sequence $\{a_n\}$ is said to converge to a real number L, if for every ϵ, there exist positive integer N such that

   A. $|a_n - L| \leq \epsilon, \forall\, n > N.$

   B. $|a_n - L| > \epsilon, \forall\, n > N.$

   C. $|a_n - L| < \epsilon, \forall\, n > N.$

   D. $|a_n - L| > \epsilon, \forall\, n > N.$

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2. Evaluate the $\lim_{n \to -2} \dfrac{n^2 + 5n + 6}{n + 2}$

   A. 0

   B. -2

   C. 1

   D. Does not exist

3. The series $\sum_{k=1}^{\infty} \left(\dfrac{\pi}{4}\right)^k$ converges to

   A. 0

   B. $\dfrac{\pi}{4 - \pi}$

   C. $\dfrac{\pi^2}{16 - \pi}$

   D. $\dfrac{\pi^2}{4(4 - \pi)}$

4. If a sequence $\{a_n\}$ is said to be **monotonically increasing**, then

   A. $a_n < a_{n+1}, \forall\, n \in N$

   B. $a_n \leq a_{n+1}, \forall\, n \in N$

   C. $a_n > a_{n+1}, \forall\, n \in N$

   D. $a_n \geq a_{n+1}, \forall\, n \in N$

5. The sum of the geometric series $2 - 1 + \dfrac{1}{2} - \dfrac{1}{4} + \dfrac{1}{8} \cdots$ is

   A. $\dfrac{4}{3}$

   B. $\dfrac{5}{4}$

C. 1

D. 3

6. Which of the following sequences converges?

A. $\left\{n + \frac{3}{n}\right\}$.

B. $\left\{-1 + \frac{(-1)^n}{n}\right\}$.

C. $\left\{\frac{\sin n\pi}{2}\right\}$.

D. $\left\{\frac{n}{\ln n}\right\}$.

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7. Which of the following is not true?

A. Every convergent is bounded

B. Every bounded monotonic sequence is convergent.

C. Every bounded sequence is convergent.

D. The limit of a sequence is unique.

8. Calculate $\lim_{n\to\infty} \frac{\ln n}{n}$

A. 0.

B. 9.

C. 1.

D. ∞.

9. The sequence $\{a_n\}$, where $a_n = \frac{6n-1}{n+3}$ is

A. increasing and bounded

B. decreasing and bounded

C. increasing and unbounded

D. decreasing and unbounded

10. A sequence $\{u_n\}$ is defined by the recursion formula $u_{n+1} = \sqrt{3u_n}$, $u_1 = 1$. Find $\lim_{n\to\infty} u_n$

A. $\sqrt{3}$

B. ∞

C. 0

D. 3

11. Find $\lim_{n\to\infty} (21^n + 76^n)^{\frac{1}{n}}$

A. 76

B. 21

C. ∞

D. 0

12. Find the sum to infinity of the geometric series $\sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n$

A. $\frac{1}{1}$

B. $\frac{1}{2}$

C. $\frac{1}{3}$

D. ∞

13. Find the sum of the convergent series $\sum_{n=0}^{\infty} 9^{-n+2}4^{n+1}$

$S_\infty = 257.143$

14. Evaluate $\sum_{n=0}^{\infty} \frac{2}{n(n+1)}$

$\sum_{n=0}^{\infty} \frac{2}{n(n+1)} = 1.0$

15. Classify the sequence $\left\{\frac{n+1}{n}\right\}$ as monotonic increasing or decreasing?

monotonic increasing sequence

16. The sequence $\left\{\frac{n}{n+1}\right\}$ is monotonic decreasing?

A. False

B. True

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17. Evaluate $\lim_{n\to\infty} \sqrt{\frac{4-2n+2n^2}{4n^2+5n}}$

A. $\sqrt{\frac{1}{2}}$

B. 1

C. $\sqrt{2}$

D. $\sqrt{\frac{2}{5}}$

18. Every convergent sequence has —— one limit point.
    A. at least
    B. at most
    C. exactly
    D. none of the above

19. The series $1^3 + 2^3 + 3^3 + \cdots$ is
    A. Divergent
    B. Convergent
    C. Bounded
    D. none of the above

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20. The series $\sum_{n=1}^{\infty} \frac{n^2}{3^n}$ is
    A. Divergent
    B. Convergent

END OF MID-SEMESTER EXAMINATION